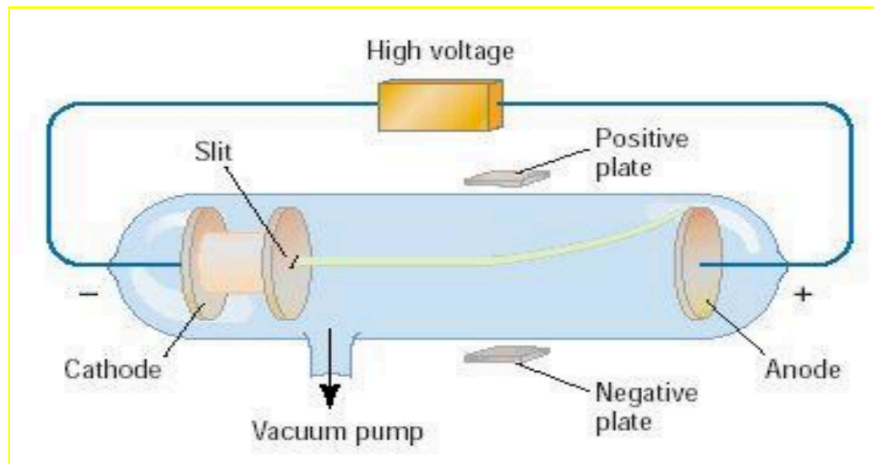


Atomic Structure

Discovery of Electron

Discovery of the Electron

In 1897, J.J. Thomson used a *cathode ray tube* to deduce the presence of a negatively charged particle: the **electron**



Conclusions from the Study of the Electron

- Cathode rays have identical properties regardless of the element used to produce them.
- All elements must contain identically charged electrons.
- Atoms are neutral, so there must be positive particles in the atom to balance the negative charge of the electrons.
- Electrons have so little mass that atoms must contain other particles that account for most of the mass.

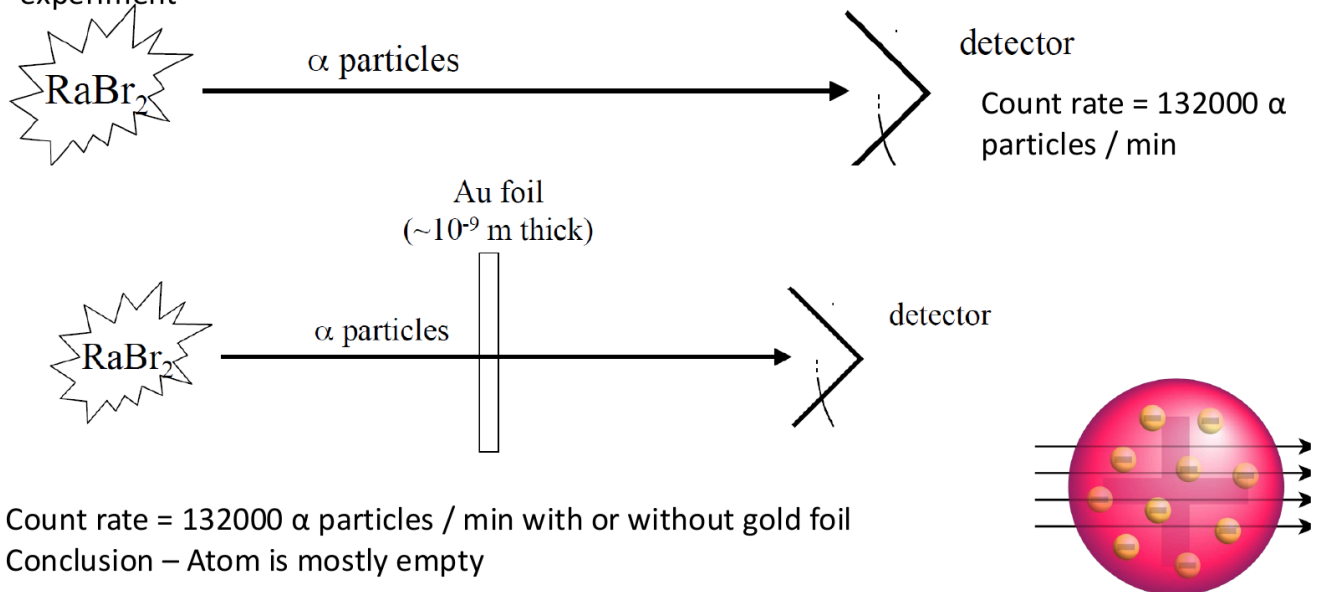
Thomson's Plum Pudding Model

Negative charges embedded within larger spheres of positive charge.

Discovery of the Nucleus

- Ernest Rutherford (Cambridge University, England, 1871-1937) studied α emission from newly-discovered radioactive elements that he got from his friend Marie Curie.
- Radium Bromide Shipped to Rutherford for experiment from France to England

With postdocs Hans Geiger and undergraduate E. Marsden, the Rutherford Lab did the experiment



The Experiment

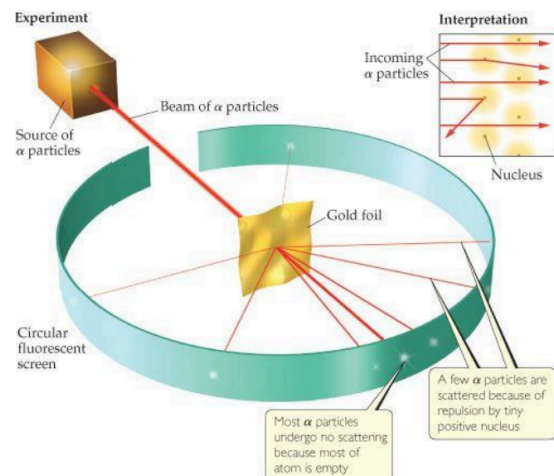
Little advance experiment from the Rutherford Laboratory: Now the detector is all around for detection of back scatter alpha particles

Mass of alpha particle: 8000 electron mass
Speed: 2×10^7 m/s

Count rate on backscatter side = 20 α particles / min

Probability (P) of backscattering:

$$P = \frac{\text{count rate backscattered}}{\text{count rate of incident particles}} = \frac{20}{132000} = 2 \times 10^{-4} \text{ (0.02\%)}$$



Conclusion

- The gold atoms are mostly empty.
- A very tiny percentage of α particles hit something massive in the atom and backscatter (bounce back). This observation indicates that most of the mass of the atom is concentrated in a very small volume relative to the volume of the entire atom. We now call this the NUCLEUS.
- Rutherford proposed the charge on the nucleus to be positive, since electrons are negatively charged and atoms are neutral. Charge of the electrons in an atom = $-Ze$. Z = atomic number e = absolute value of electron's charge. Charge of the nucleus of an atom = positive = $+Ze$

- Rutherford calculated the diameter of the nucleus to be $10^{-14}m$. His calculation related the probability of backscattering to the diameter of the nucleus based on the size of the atom and the thickness of the foil.

Rutherford Scattering formula (Annexure of Chapter 4 – PLEASE DO. It will be taught in detail by Prof. Shandilya)

The formula that Rutherford obtained for alpha particle scattering by a thin foil on the basis of the nuclear model of the atom is

Rutherford Scattering Formula

$$N(\theta) = \frac{N_i n t Z^2 e^4}{(8\pi\epsilon_0)^2 r^2 K E^2 \sin^4(\frac{\theta}{2})}$$

$N(\theta)$ = number of alpha particles per unit area that reach the screen at a scattering angle of θ

N_i = total number of alpha particle that reach the screen

n = number of atoms per unit volume in the foil

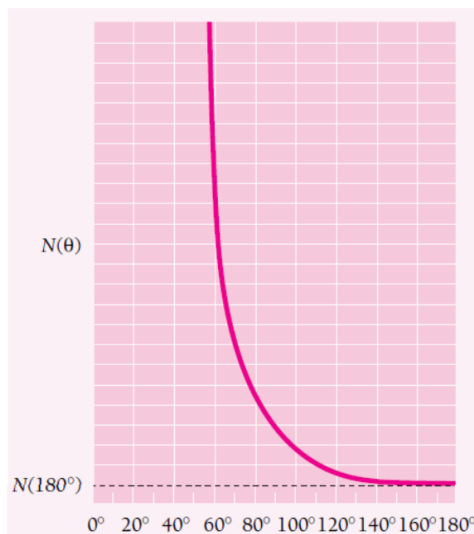
Z = atomic number of the foil atoms

r = distance of the screen from the foil

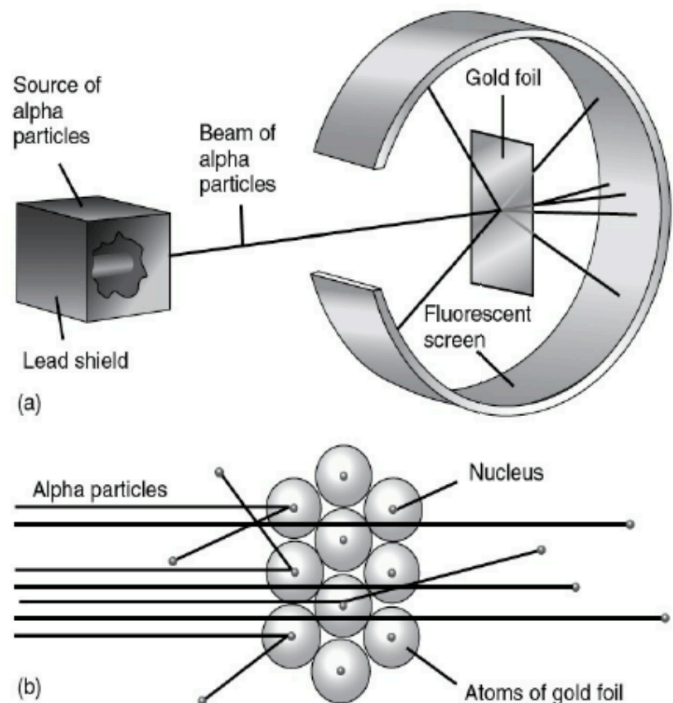
KE = kinetic energy of the alpha particles

t = foil thickness

The predictions of Scattering Formula agreed with the measurements of Geiger and Marsden, which supported the hypothesis of the nuclear atom.



Because $N(\theta)$ is inversely proportional to $\sin^4(\theta/2)$ the variation of $N(\theta)$ with θ is very pronounced : only 0.14 percent of the incident alpha particles are scattered.



Nuclear Dimensions

Rutherford assumed $\rightarrow$ the size of a target nucleus $<$ the minimum distance R (the distance the incident alpha particles approach the nucleus before being deflected away)

R = The distance of **closest approach** R for the most energetic alpha particles.

Rutherford scattering $\rightarrow$ gives us a way to find an upper limit to nuclear dimensions.

An alpha particle will have its **smallest R** when **it approaches a nucleus head on**, which will be followed by a **180° scattering**.

At the instant of closest approach the initial kinetic energy (KE) of the particle is entirely converted to electric potential energy, and so at that instant

$$KE_{initial} = PE = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{R}$$

since the charge of the alpha particle is $2e$ and that of the nucleus is Ze .

Distance of closest approach

$$R = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{KE_{initial}}$$

The maximum KE found in alpha particles of natural origin is 7.7 MeV,

$$7.7 \text{ MeV} = 1.2 \times 10^{-12} \text{ J}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$$

$$\begin{aligned} R &= \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{KE_{initial}} \\ &= \frac{9 \times 10^9 \text{ Nm}^2 (2Z)(1.6 \times 10^{-19} \text{ C})^2}{\text{C}^2 \frac{1.2 \times 10^{-12} \text{ J}}{}} \\ &= 3.8 \times 10^{-16} Z \text{ m} \end{aligned}$$

The atomic number of gold is $Z=79$

$$R = 3.0 \times 10^{-14} \text{ m}$$

Radius of the gold nucleus $< 3.0 \times 10^{-14} \text{ m} \rightarrow$ less than 10^{-4} times of the atomic radius

Electron Orbits

ELECTRON ORBITS

Hydrogen atom : Single electron.

Assume a circular electron orbit for convenience.

The centripetal force

$$F_c = \frac{mv^2}{r}$$

holding the electron in an orbit r from the nucleus is provided by the electric force between them

$$F_e = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2}$$

The condition for a dynamically stable orbit is

$$F_c = F_e$$

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2}$$

$$\text{Electron velocity, } v = \frac{e}{\sqrt{4\pi\epsilon_0 mr}}$$

The total energy E of the electron in a hydrogen atom is the sum of its kinetic and potential energies, which are

$$KE = \frac{1}{2}mv^2 \quad PE = -\frac{1}{4\pi\epsilon_0} \frac{e^2}{r}$$

The minus sign $\rightarrow$ $PE = 0$ at $r = \infty$, that is, when the electron and proton are infinitely far apart

$$E = KE + PE = \frac{1}{2}mv^2 - \frac{1}{4\pi\epsilon_0} \frac{e^2}{r}$$

$$E = \frac{1}{8\pi\epsilon_0} \frac{e^2}{r} - \frac{1}{4\pi\epsilon_0} \frac{e^2}{r}$$

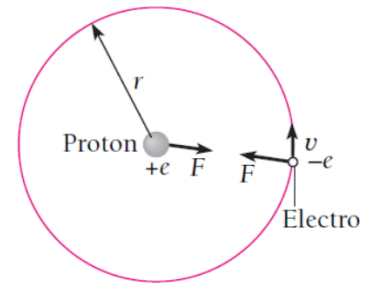
Total energy of hydrogen atom

$$E = -\frac{1}{8\pi\epsilon_0} \frac{e^2}{r}$$

The total energy of the electron is negative.

This holds for every atomic electron and reflects the fact that it is bound to the nucleus.

If $E > \text{zero}$ $\rightarrow$ an electron would not follow a closed orbit around the nucleus.



(FAILURE OF) THE CLASSICAL DESCRIPTION OF AN ATOM

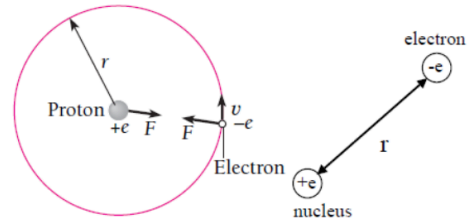
How is an electron bound to the nucleus? What is the force between them? **Coulomb's Force Law** describes the force of attraction and repulsion between charged particles.

$$\text{Coulomb Force } (F_r) = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2}$$

Q_x = charge on particle x

r = distance between two charges

ϵ_0 = permittivity constant of a vac
($8.854 \times 10^{-12} \text{ C}^2 \text{J}^{-1} \text{m}^{-1}$)



When charges are the same sign, acceleration of the particles will push them apart. The force is positive and repulsive.

When charges are the opposite sign, acceleration will pull them together. The force is negative and attractive.

Consider an H atom ($Z=1$) with 1 electron and 1 proton.
When $r \rightarrow \infty$, $F(r) = 0$

If the electron and the nucleus are infinitely attracted, why do they stay apart.

As $r \rightarrow 0$, $F(r) = \infty \rightarrow$ infinitely attractive

Coulomb's law only describes force as a function of r (distance). It doesn't tell us how r changes with time, t .

Newton's Second law:

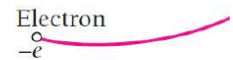
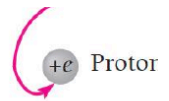
Force, $F = ma$ (mass X acceleration)

$F = m (dv/dt)$ (function of velocity) = $m(d^2r/dt^2)$ (function of distance)

Plug in Coulomb's force law

For $r_{\text{initial}} = 0.5 \times 10^{-10} \text{m}$, calculate how fast it should go to zero. The answer is very fast.

$r=0$ at $t = 10^{-10} \text{ s}$



This predicts that the electron should crash into the nucleus in 10^{-10} sec. But why is it not doing that?

The laws of classical Mechanics no longer work at this size scale.

Atomic Spectra and Bohr's Model

- Bohr's postulates were based on the observation of discrete spectra and the results of the α -scattering experiment of Rutherford.
- Electrons move around the nucleus in discrete orbits.
- These orbits correspond to fixed energies and are stable (i.e. no radiation is emitted by such electrons)
- Radiation is emitted only when the electron jumps from one discrete orbit to another orbit of a lower energy. When electrons absorb radiation, the reverse transition occurs.
- A dynamic equilibrium exists between the centrifugal force and the coulomb force between the electron and the nucleus.

$$\frac{(Ze)^2}{4\pi\epsilon_0 r^2} = m\omega^2 r \rightarrow \omega = \frac{e^2}{4\pi\epsilon_0 m r^3}$$

The electron revolves at the frequency ω in the orbit. Classical electrodynamics predicts that the charge will radiate at this very frequency and lose energy. Bohr's postulates override this.

The total energy of the electron is

$$\frac{(Ze)^2}{4\pi\epsilon_0 r^2} = m\omega^2 r \rightarrow \omega = \frac{e^2}{4\pi\epsilon_0 m r^3}$$

$$E = E_{kin} + E_{pot} = \frac{1}{2} m \omega^2 r^2 - \frac{e^2}{4\pi\epsilon_0 r}$$

Using the above obtained value of ω , we get

$$E = -\frac{e^2}{8\pi\epsilon_0 r} = \frac{1}{2} \frac{\omega^{2/3}}{(4\pi\epsilon_0)^{2/3}} (me^4)^{1/3}$$

Bohr's important insight was to recognize that the frequency of radiation (according to the postulates) should be given by

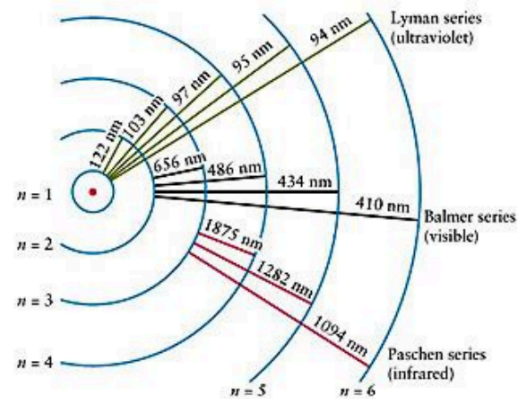
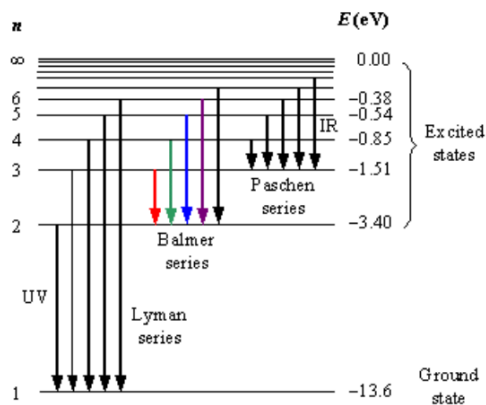
$$h\nu = E_n - E_{n'}$$

where $h\nu$ is the radiation quantum as proposed by Planck and that this radiation must be the same as that.

Balmer series - wavelengths in the visible portion of the hydrogen spectrum.

Lyman series - wavelengths in the ultraviolet region > smallest is 91.2 nm > 13.598eV

Rydberg constant = $10 \times 10^7 m^{-1}$ $1.097 \times 10^7 m^{-1}$



Hydrogen Series of Spectral Lines $\frac{1}{\lambda} = R \left(\frac{1}{n^2} - \frac{1}{k^2} \right)$

Discoverer (year)	Wavelength	n	k
Lyman (1916)	Ultraviolet	1	>1
Balmer (1885)	Visible, ultraviolet	2	>2
Paschen (1908)	Near Infrared	3	>3
Brackett (1922)	Infrared	4	>4
Pfund (1924)	Far Infrared	5	>5

Rydberg-Ritz Combination Principle

The wavenumber of any spectral line in an atom's spectrum can be expressed as the sum or difference of the wavenumbers of two other spectral lines from the same element

THE BOHR ATOM

The de Broglie wavelength of an electron is $\lambda = \frac{h}{mv}$

v = electron velocity

$$v = \frac{e}{\sqrt{4\pi\epsilon_0 mr}}$$

Orbital electron wavelength

$$\lambda = \frac{h}{e} \sqrt{\frac{4\pi\epsilon_0 r}{m}}$$

Radius of electron orbit, $r = 5.3 \times 10^{-11} \text{ m}$
 The electron wavelength $\lambda = \frac{6.63 \times 10^{-34} \text{ J} \cdot \text{s}}{1.6 \times 10^{-19} \text{ C}} \sqrt{\frac{(4\pi)(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(5.3 \times 10^{-11} \text{ m})}{9.1 \times 10^{-31} \text{ kg}}}$
 $= 33 \times 10^{-11} \text{ m}$

This wavelength is exactly the same as the circumference of the electron orbit,

$$2\pi r = 33 \times 10^{-11} \text{ m}$$

Only waves with wavelengths that just fit in survive (all others cancel themselves out)

Bohr's Orbit: An electron can circle a nucleus only if its orbit contains an integral number of de Broglie wavelengths.

Condition for orbit stability

$$n\lambda = 2\pi r_n \quad n = 1, 2, 3 \dots$$

r_n = the radius of the orbit that contain n wavelengths.

n = the **quantum number** of the orbit.

Substituting for λ , the electron wavelength

$$\frac{nh}{e} \sqrt{\frac{4\pi\epsilon_0 r_n}{m}} = 2\pi r_n$$

Orbital electron wavelength $\lambda = \frac{h}{e} \sqrt{\frac{4\pi\epsilon_0 r}{m}}$

The possible electron orbits are those whose radii



$$r_n = \frac{n^2 h^2 \epsilon_0}{\pi m e^2} \quad n = 1, 2, 3, \dots$$

The innermost radius is called Bohr Radius

Bohr radius

$$a_0 = r_1 = 5.292 \times 10^{-11} \text{ m}$$

The other radii in terms of Bohr Radius are

$$r_n = n^2 a_0$$

ENERGY LEVELS AND SPECTRA

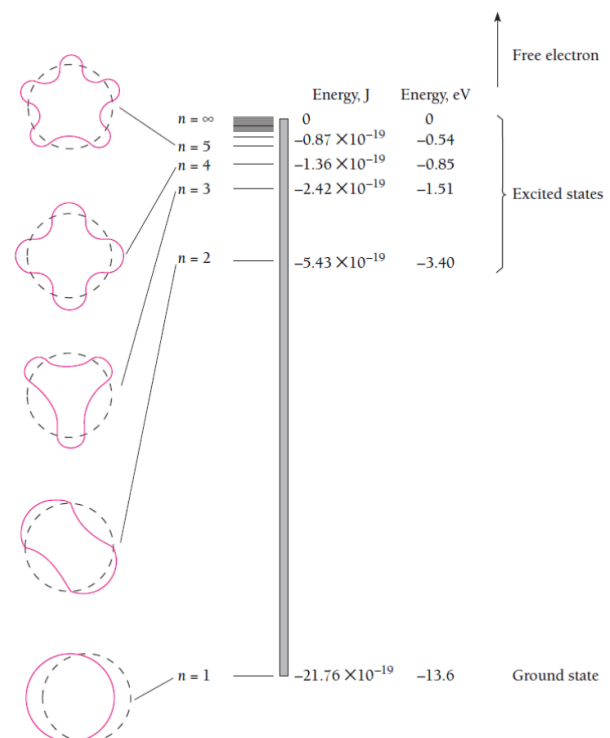
The electron energy E_n is given in terms of the orbit radius r_n as

$$E_n = -\frac{e^2}{8\pi\epsilon_0 r_n}$$

The **energy levels** of the hydrogen atom

$$E_n = -\frac{me^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n^2} \right) = \frac{E_1}{n^2} \quad n = 1, 2, 3, \dots$$

$$E_1 = -2.18 \times 10^{-18} \text{ J} = -13.6 \text{ eV}$$



Bohr's Correspondence Principal

approaches classical mechanics.

In limit that $n \rightarrow \infty$,

quantum mechanics must agree with classical physics

$$E_{\text{photon}} = 13.6 \text{ eV} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) = hf_{\text{photon}}$$

$$n_i \approx n_f \text{ when both } n_i, n_f \gg 1$$

Bohr's correspondence principle considers transitions between **nearby very-high- n states**, for example

$$n_i = n, n_f = n - 1,$$

with

$$n \rightarrow \infty.$$

Although n and $n - 1$ always differ by exactly 1, their **relative difference**

$$\frac{n_i - n_f}{n_i} = \frac{1}{n}$$

goes to zero as $n \rightarrow \infty$.

For very large n , we can approximate

$$\frac{1}{(n-1)^2} \simeq \frac{1}{n^2} \left(1 + \frac{2}{n} \right),$$

so

$$\frac{1}{(n-1)^2} - \frac{1}{n^2} \simeq \frac{2}{n^3}.$$

Therefore,

$$hf \simeq \frac{27.2 \text{ eV}}{n^3}.$$

The beautiful part of the correspondence principle $\rightarrow$ If you calculate the **classical orbital frequency** of the electron in the n -th Bohr orbit,

$$f_{\text{orbit}} = \frac{v_n}{2\pi r_n},$$

v_n = speed of the electron in the n -th Bohr orbit.

you obtain exactly the same large- n result as the photon frequency:

$$f_{\text{photon}} \rightarrow f_{\text{orbit}}$$

for

$$n \rightarrow \infty, \Delta n = 1.$$

For very large quantum numbers, transitions between neighbouring quantum states produce radiation whose frequency approaches the classical orbital frequency of the electron.

This is Bohr's correspondence principle:

Quantum physics $\rightarrow$ classical physics in the appropriate large-quantum-number limit.

$n = 1, 2, 3$	:	levels clearly discrete
$n = 100, 101, 102$	:	levels extremely close
$\downarrow$		
$n \rightarrow \infty$	:	almost a continuum $\rightarrow$ classical behaviour

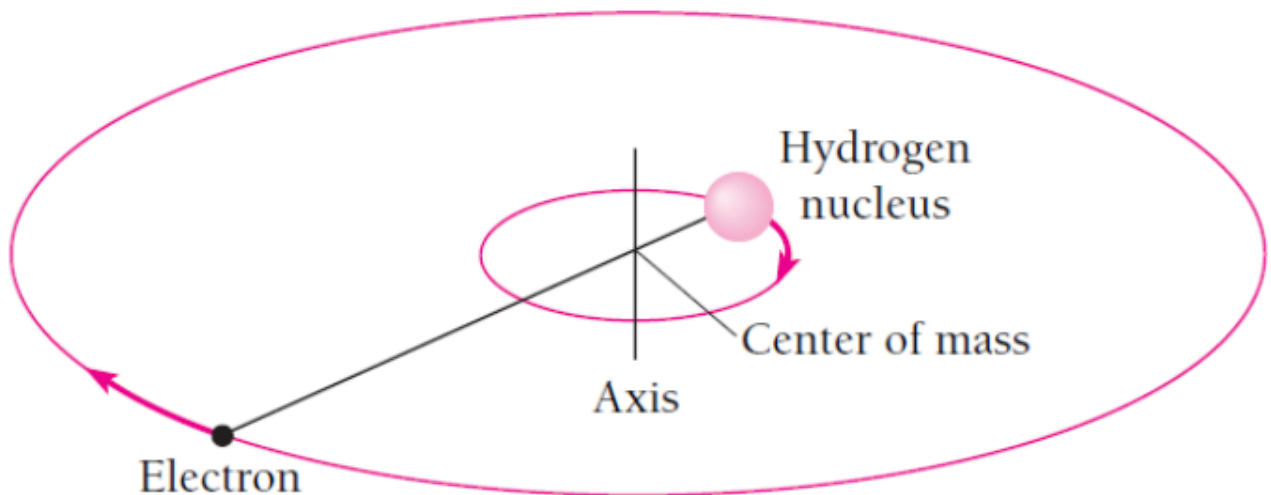
Conclusion: Spectroscopy provides direct evidence for **quantization of energies** (electronic, vibrational, rotational etc.) of atoms and molecules

Limitations of Bohr Model & Rydberg-Ritz Equation

- The model only works for hydrogen (and other one electron ions) – **ignores e-e repulsion**.
- Does not explain **fine structure** of spectral lines.
- Note:** The Bohr model (assuming circular electron orbits) is fundamentally incorrect.

Nuclear Motion

in reality, both the nucleus and electron orbit their common center of mass, which lies near the nucleus due to its much greater mass. This system is equivalent to a single particle of mass m' orbiting the heavier nucleus.



Reduced mass

$$m' = \frac{mM}{m + M}$$

m = electron mass
 M = nuclear mass

To account for the nucleus's motion in a hydrogen atom, we replace the electron with a particle of mass m' . The atom's energy levels then become

Energy levels corrected for nuclear motion

$$E'_n = -\frac{m'e^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n^2}\right) = \left(\frac{m'}{m}\right) \frac{E_1}{n^2}$$

$$E_n = -\frac{me^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n^2}\right) = \frac{E_1}{n^2}$$

$$-\frac{e^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n^2}\right) = \frac{E_1}{mn^2}$$

$$E_1 = -\frac{e^4}{8\epsilon_0^2 h^2}$$

Due to the nucleus's motion, all hydrogen energy levels shift by the fraction

$$\frac{m'}{m} = \frac{M}{M + m} = 0.99945$$

This represents an increase of 0.055 percent because the energies E_n , being smaller in absolute value, are therefore less negative.

Energy levels
corrected for
nuclear motion

$$E'_n = -\frac{m'e^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n^2} \right) = \left(\frac{m'}{m} \right) \left(\frac{E_1}{n^2} \right)$$

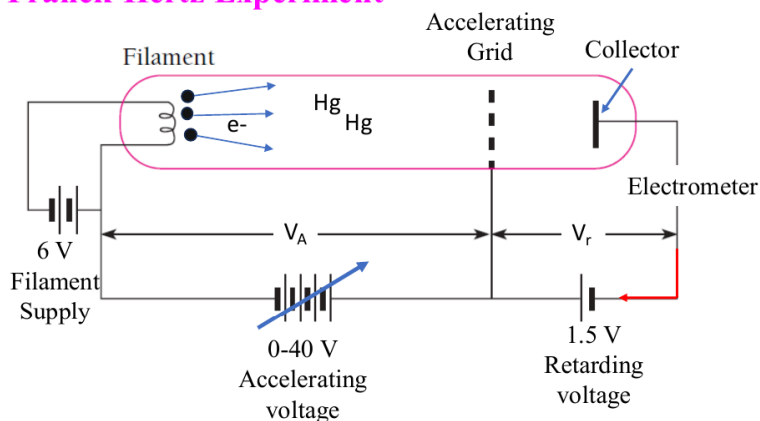
H atom $E = -13.606 \text{ eV}$ $\rightarrow$ -13.598 eV $\leftarrow$ reduced mass

Experiment $E = -13.598 \text{ eV}$

- The energy equation removes the discrepancy between the predicted wavelengths of the spectral lines of hydrogen and the measured ones.
- The value of the **Rydberg constant R** without correcting for nuclear motion is $1.0973731 \times 10^7 \text{ m}^{-1}$;
- The correction lowers **R** to $1.0967758 \times 10^7 \text{ m}^{-1}$.
- The notion of reduced mass played an important part in the discovery of deuterium,
- Deuterium nucleus contains a neutron as well as a proton.
- Because of the greater nuclear mass, the spectral lines of deuterium are all shifted slightly to wavelengths shorter than the corresponding ones of ordinary hydrogen.
- H α line of deuterium > Transition $n = 3 > n = 2$ energy level > Wavelength 656.1 nm,
- H α line of Hydrogen > Transition $n = 3 > n = 2$ energy level > Wavelength 656.3 nm.
- Identification of deuterium in 1932 by the American chemist Harold Urey.

Frank-Hertz Experiment

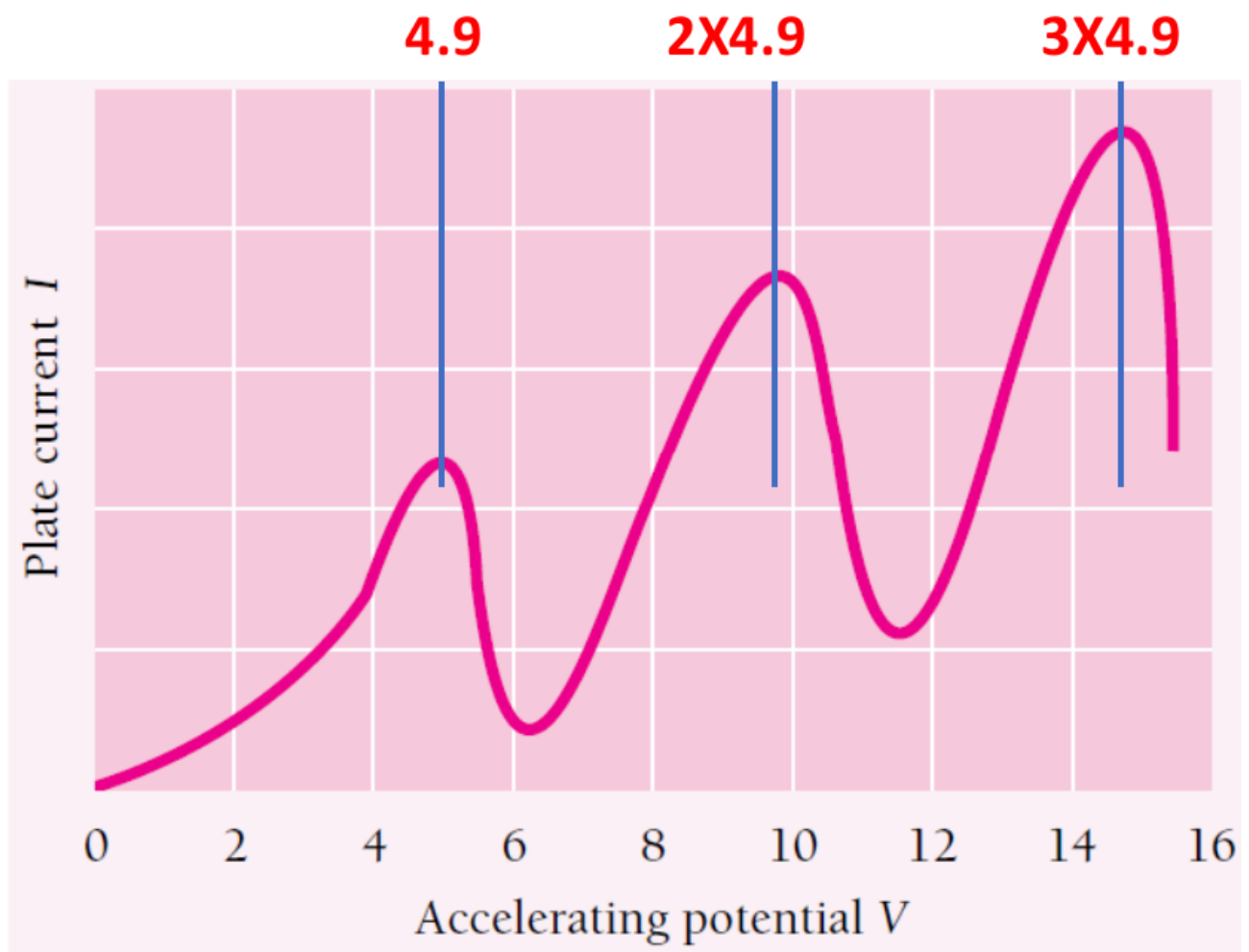
Franck-Hertz Experiment



- The Filament ejects electrons in the gas filled tube.
- A variable (positive) accelerating voltage, V_A , is applied to the grid.
- The electron acquires kinetic energy $K = eV_A$

- There is (negative) retarding voltage (V_r) between the grid and collector.
- Only electrons having enough energy will overcome this potential and reach the collector and contribute to current measured with an Electrometer.

- If $V_r > V_A \rightarrow$ no electron can reach the collector, so no current would be measured.
- If $V_r < V_A \rightarrow$ if the tube is highly evacuated, most of the electrons would reach the collector and have energy $|e|(V_A - V_r)$
- Franck and Hertz observed the collector current as a function of $V_A (> V_r)$ when tube was filled with various gases (result for mercury gas is shown here)
- At first, the current increased as was expected for a typical vacuum tubes, **but as $\sim 4.9\text{V}$ current suddenly dropped.**
- Then, the increase resumed until 9.8 V , and so on.



- The current drops because fewer electrons reach the collector.
- This occurs only if the electrons undergo elastic collisions.
- Thus, when $V_A = 4.9 \times n$ Volts ($n=1, 2, \dots$) the electrons undergo inelastic collisions with Hg atoms.
- In inelastic collision kinetic energy of electrons is transferred to the internal energy of Hg atoms. Hg atoms absorb the energy of electrons.

